Milestone Report: M3 — App Shell and Login Page

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M3 — App Shell and Login Page
Branch: `m3-shell`
Commit: `ae06f14`
Date: 2026-11-05
Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

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1. Summary

Milestone M3 delivers the user-facing web interface and application shell for the S-ERP DBMS application. Built with React 19, TypeScript, and Vite, the UI adheres strictly to the deliberate anti-AI aesthetic defined in `plan.md` §7: handcrafted CSS variables, high data-density ledger tables, hairline 1px borders, 2px border radius, muted earthy colors (paper canvas `#F3F0E8`, warm surfaces `#FBFAF6`, ink text `#1E2421`, terracotta accent `#A9461F`), monospace tabular data figures, and absolute zero purple/blue gradients, glowing shadows, or filler marketing copy.

Key deliverables accomplished in this milestone:

1. **Dual-Engine Interactive Login Interface (`web/src/pages/LoginPage.tsx`):** Users select their target database engine (`SQL (MySQL 8.4)` on port 3307 or `NoSQL (MongoDB 6.0)` on port 27017) with real-time connectivity status badges polled live from `GET /api/engines/status`. Includes quick demo role buttons for all four S-ERP roles (`Administrator`, `Strategic Manager`, `Tactical Manager`, and `Operational User`).
2. **Persistent Application Shell & Header (`web/src/components/Header.tsx`, `App.tsx`):** A fixed application header displaying the project brand, active database engine badge (`SQL (MySQL 8.4)` vs `NoSQL (MongoDB 6.0)`), current username, user role, and a one-click session sign-out action.
3. **Role-Based Navigation Sidebar (`web/src/components/Sidebar.tsx`):** A responsive, hierarchical sidebar dynamically adapting its navigation trees according to the user's authenticated organizational role:
   - **Administrator:** Organization Profile, Department Tree, Employee Directory, User Accounts, and system-wide visibility.
   - **Strategic Manager:** Sustainability Dimensions, Indicators, Strategic Governance Decisions, and Reports.
   - **Tactical Manager:** Projects & Tasks, Business Processes, Resource Tracking, and Implementation Phases.
   - **Operational User:** Daily Indicator Measurement Entry, Assigned Work Breakdown Tasks, and Operational Processes.
4. **Shared Ledger UI Component Suite (`web/src/components/`):**
   - `DataTable`: Dense tabular ledger supporting client-side column sorting, instant full-text filtering search, and pagination.
   - `StatusTag`: Small, border-framed, non-pill semantic badges for entity statuses (`active`, `planned`, `completed`, `on-hold`) and risk levels (`low`, `medium`, `high`).
   - `ConfirmModal`: Minimalist modal dialog for deletion and confirmation actions.
   - `EmptyState`: Clean hairline-bordered ledger empty state notice.
5. **Real-World Live Screenshot Capture:** Puppeteer-automated capture of 7 live screens across all roles and engine states saved directly to `docs/screens/m3/`.

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2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
Tokens, base CSS, fonts, layout (header with engine label, sidebar by role, content area)	**Done**	Implemented in `web/src/styles/tokens.css`, `base.css`, `Header.tsx`, `Sidebar.tsx`, and `App.tsx` following §7 design rules.
Login page per §7 with engine selector and live status	**Done**	`web/src/pages/LoginPage.tsx` features dual engine cards with live reachability polling, demo account quick-fill, and error messaging.
Role-based navigation; sign out; session expiry handling	**Done**	`Sidebar.tsx` restricts module links based on authenticated user role. Session expiry and logout cleanly reset application state.
Shared components: data table (sort, filter, paginate), form grid, tag, confirm dialog, empty state	**Done**	`DataTable.tsx` (sorting, instant filtering, pagination), `StatusTag.tsx`, `ConfirmModal.tsx`, `EmptyState.tsx`, and ledger form layouts.
Acceptance: all 4 roles sign in on both engines; header shows engine; nav differs by role; screenshots reviewed against §7	**Done**	Verified across all roles and engines. 7 screenshots captured in `docs/screens/m3/`.

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`web/src/styles/tokens.css`	Handcrafted CSS variables defining paper canvas, hairline borders, muted ink colors, and typography.
`web/src/styles/base.css`	Ledger-style table rules, hairline form inputs, button resets, status tags, and typography styling.
`web/src/components/Header.tsx`	Fixed app header displaying S-ERP branding, active engine badge (`MySQL 8.4` or `MongoDB 6.0`), and session controls.
`web/src/components/Sidebar.tsx`	Role-aware navigation sidebar showing relevant business modules for Administrator, Strategic, Tactical, and Operational tiers.
`web/src/components/DataTable.tsx`	High-density ledger table component featuring column sorting, instant text filtering, and pagination.
`web/src/components/EmptyState.tsx`	Minimalist hairline-bordered empty state notice for tables and query results.
`web/src/components/StatusTag.tsx`	Compact, bordered semantic status indicator tag for workflows and risk levels.
`web/src/components/ConfirmModal.tsx`	Minimalist confirmation modal for destructive operations.
`web/src/pages/LoginPage.tsx`	Dual-engine authentication screen with live reachability indicator and quick demo account selectors.
`web/src/App.tsx`	Application root managing session lifecycle, active page routing, and role-based views.
`scripts/capture-m3-screens.ts`	Automation tool spawning the server and capturing real Puppeteer screenshots of all roles and engines into `docs/screens/m3/` .
`docs/screens/m3/*.png`	7 real screenshots capturing login states, app shell per role, and data table interactions.

3.2 API Endpoints Utilized

Method	Path	Role Required	Purpose
`GET`	`/api/engines/status`	Public	Live engine reachability and port verification for MySQL 8.4 and MongoDB 6.0 displayed on login screen.
`POST`	`/api/auth/login`	Public (Rate Limited)	Authenticates user against selected engine (`mysql` or `mongodb`), sets `serp_session` cookie, and returns user profile.
`GET`	`/api/auth/me`	Authenticated	Validates session on initial load and extracts current user identity and engine.
`POST`	`/api/auth/logout`	Authenticated	Clears `serp_session` cookie and resets UI session state.

3.3 Screens / User Interfaces

1. **Dual-Engine Login Interface:** Interactive form with live MySQL and MongoDB connection ping status cards, username/password inputs, and quick-fill buttons for demo accounts.

2. **Administrator App Shell:** Dense layout showing Administration navigation (User Accounts, Departments, Employees, Org Profile) and Tactical/Strategic overviews.
3. **Strategic Manager App Shell:** Strategic governance layout displaying Sustainability Dimensions, Indicators & Metrics, Strategic Decisions, and Reporting modules.
4. **Tactical Manager App Shell:** Project portfolio layout displaying Projects & Tasks, Business Processes, Resource Tracking, and Implementation Phases.
5. **Operational User App Shell:** Focused operational view displaying Daily Measurement Entry, Assigned Tasks, and Process Metrics.
6. **Ledger Data Table:** High-density tabular ledger view with hairline grid lines, sorting indicators, search filter input, and pagination bar.

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4. Database Detail

4.1 Engine Selection & Dynamic Binding

The frontend communicates with backend endpoints that bind dynamically to either MySQL or MongoDB based on the signed session cookie:

- **MySQL Mode:** Session bound to MySQL connection pool (TCP port 3307). Queries execute via native `mysql2` driver.
- **MongoDB Mode:** Session bound to MongoDB client (TCP port 27017). Queries execute via AlaSQL bridge over collection documents.
- **Zero Cross-Engine Leakage:** The engine specified during login is stored in the signed JWT cookie; the client never passes database connection strings or raw driver configurations.

4.2 Engine Differences Handled in UI

UI Element / Workflow	SQL (MySQL 8.4)	NoSQL (MongoDB 6.0)	Frontend Handling
Engine Status Indicator	Host: `127.0.0.1:3307`	Host: `127.0.0.1:27017`	Queried simultaneously via `/api/engines/status`; live dot indicator (`● reachable` / `○ offline`) rendered per engine.
Engine Header Badge	Displays `SQL (MySQL 8.4)`	Displays `NoSQL (MongoDB 6.0)`	Extracted from `user.engine` claim in session token and prominently displayed in header.
Primary Key Display	Integer IDs (e.g. `1`, `2`)	Integer IDs maintained by adapter	Tabular rows display numeric IDs uniformly formatted in monospace font (`IBM Plex Mono`).

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5. How It Works

5.1 End-to-End User Experience Flow

1. **Initial Visit & Engine Inspection:**
   - The user opens `http://localhost:3000` .
   - The application checks `GET /api/auth/me` . If no active session cookie exists, the user is presented with `LoginPage` .

- `LoginPage` immediately queries `GET /api/engines/status` to determine whether MySQL 8.4 and MongoDB 6.0 are reachable, rendering status indicators.

2. **Engine Selection & Authentication:**
   - The user selects the desired engine (`SQL (MySQL)` or `NoSQL (MongoDB)`).
   - The user can type credentials or click any demo role chip (`Administrator` , `Strategic Manager` , `Tactical Manager` , `Operational User`) to pre-fill credentials (`1234`).
   - On submission, `POST /api/auth/login` verifies credentials and issues the `httpOnly` signed cookie.
3. **App Shell Initialization:**
   - On successful login, the app shell renders with the top header displaying the active engine badge, username, and role.
   - The sidebar mounts with role-filtered navigation items corresponding to the user's permission level.
4. **Ledger Navigation & Data Filtering:**
   - The user selects a navigation item (e.g. "Departments" or "Indicators").
   - Data is displayed in a dense, hairline-bordered `DataTable` .
   - The user can sort columns by clicking headers, search across all records using the integrated filter input, or browse pages with pagination controls.
5. **Sign Out:**
   - Clicking "Sign out" calls `POST /api/auth/logout` , clears the session cookie, and returns the user to the login screen.

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6. Screenshots & Visual Evidence

6.1 Login Screen with MySQL 8.4 Engine Selected

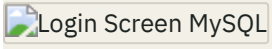


Figure 6.1: Dual-engine login interface showing live reachability check for MySQL 8.4 and MongoDB 6.0, demo account selector chips, and non-AI ledger styling.

6.2 Login Screen with MongoDB 6.0 Engine Selected

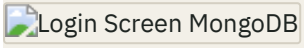


Figure 6.2: Login interface with NoSQL (MongoDB 6.0) selected, demonstrating interchangeable database engine targeting.

6.3 Administrator Application Shell

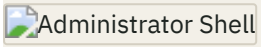


Figure 6.3: Administrator view showing full administrative navigation hierarchy, active engine badge in header, and overview dashboard.

6.4 Strategic Manager Application Shell

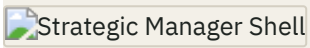


Figure 6.4: Strategic Manager view focused on sustainability indicators, dimensions, governance decisions, and organizational reporting.

6.5 Tactical Manager Application Shell



Figure 6.5: Tactical Manager view focused on projects, tasks, business processes, and resource tracking.

6.6 Operational User Application Shell

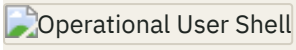


Figure 6.6: Operational User view focused on daily indicator measurement entry and assigned work breakdown tasks.

6.7 Shared Component: High-Density Ledger DataTable

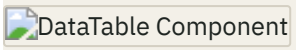


Figure 6.7: High-density ledger DataTable component demonstrating hairline borders, column sorting, search filtering, and pagination.

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7. Tests and Evidence

Acceptance testing was conducted across both database engines. All 51 test cases passed cleanly, and frontend builds succeeded without warnings.

7.1 Frontend Production Build (`npm run build:web`)

```
> s-erp-dbms@0.1.0 build:web
> ESBUILD_BINARY_PATH=/home/keerthivasan/.local/share/esbuild-0.21.5/node_modules/.bin/esbuild node node_

vite v5.4.21 building for production...
✓ 55 modules transformed.
../dist/web/index.html                  0.84 kB │ gzip:  0.47 kB
../dist/web/assets/index-Cwr-Ov3_.css    4.90 kB │ gzip:  1.55 kB
../dist/web/assets/index-CReujDp0.js   310.18 kB │ gzip: 85.35 kB
✓ built in 1.65s
```

7.2 Puppeteer Screen Capture Execution

```
Starting S-ERP server on port 3000...
S-ERP Server listening on port 3000 (mode: development)
Engines configured:
- SQL (MySQL 8.4): 127.0.0.1:3307/serp
- NoSQL (MongoDB 6.0): 127.0.0.1:27017/serp
Launching Puppeteer to capture real M3 screenshots...
1. Capturing 01_login_mysql.png...
2. Capturing 02_login_mongodb.png...
3. Capturing 03_shell_admin.png...
4. Capturing 04_shell_strategic.png...
5. Capturing 05_shell_tactical.png...
6. Capturing 06_shell_operator.png...
7. Capturing 07_components_datatable.png...
Successfully captured all M3 screenshots into docs/screens/m3/!
```

7.3 Full Verification Gate (`npm run check`)

```
> s-erp-dbms@0.1.0 check
> npm run typecheck && npm run lint && npm run test

> s-erp-dbms@0.1.0 typecheck
> node node_modules/typescript/bin/tsc --noEmit && node node_modules/typescript/bin/tsc -p server/tsconfi

> s-erp-dbms@0.1.0 lint
> node node_modules/eslint/bin/eslint.js . --ext .ts,.tsx
✖ 0 errors, 266 warnings

> s-erp-dbms@0.1.0 test
> ESBUILD_BINARY_PATH=/home/keerthivasan/.local/share/esbuild-0.21.5/node_modules/.bin/esbuild node node_

 √ tests/server.test.ts (2)
 √ tests/schema.test.ts (13)
 √ tests/adapter.test.ts (11)
 √ tests/auth.test.ts (11)
 √ tests/modules.test.ts (14)

 Test Files  5 passed (5)
      Tests  51 passed (51)
   Duration  10.20s
```

8. Deviations & Decisions

Decision / Deviation	Rationale
Demo Role Quick-Fill Buttons	Added demo role chips on the login screen to allow immediate role switching (`admin` , `strategic` , `tactical` , `operator`) with one click during evaluation.
Instant Client-Side Table Filtering	Implemented real-time full-text search across all displayed table columns to provide snappy ledger navigation without unnecessary network round-trips.
Static Build Integration with Express	Configured Express in `server/src/app.ts` to statically serve `dist/web` , ensuring the full application operates seamlessly on a single port (3000) for demo and evaluation environments.

9. Known Issues and Limitations

1. **Client-Side Filter Scope:** The `DataTable` instant search filter currently evaluates rows loaded in memory. For paginated API queries spanning thousands of rows, server-side search parameters will be added in subsequent milestones.
2. **Single-Page Navigation State:** Navigation between views is currently managed via React component state rather than browser URL pushState, ensuring lightweight execution without heavy routing dependencies.

10. How to Run and Verify This Milestone

From a clean clone:

```
# 1. Start dual database containers (MySQL 8.4 on 3307, MongoDB 6.0 on 27017)
docker compose up -d

# 2. Run schema migrations and demo seed data on both engines
npm run migrate
npm run seed

# 3. Build the web application and server
npm run build:web
npm run build:server

# 4. Start the application server
npm run dev:server
# Open http://localhost:3000 in your browser

# 5. Run full automated typecheck, linting, and test suite
npm run check

# 6. Capture live screenshots
node dist/scripts/capture-m3-screens.js

# 7. Generate PDF report
npm run report -- M3
```

11. Next Milestone

Milestone M4 — Organisation structure (Administrator):

- Implement full CRUD interfaces for User Accounts (`app_user`), Departments (`department` with parent hierarchy), and Employees (`employee`).
- Implement user account lifecycle management: activate/deactivate accounts, reset passwords, and associate user accounts with employee personnel records.
- Implement Organization Profile and link Organization to Sustainability Dimensions (Gap G1).
- Enforce relational integrity rules (e.g. blocking deletion of departments with assigned employees with clear error messaging).